
From Finite Blacklists to Freiling Symmetry

Exact avoidance thresholds, critical tournaments,
choice-sensitive colorings, and perfect safe sets

A finite–infinite research problem at the intersection of
extremal combinatorics, weak choice, measure, and topology

Research report prepared for Vladimir Reshetnikov

5 October 2026

Status and attribution. This is an AI-assisted, unrefereed research report. It proposes a problem, proves the results stated as theorems below, and separates those results from classical ingredients and conjectural extensions. The elementary infinite partition theorem is due to de Bruijn and Erdős (1951), not to this report. No exhaustive priority search has been made for the finite refinements proved here. No endorsement, collaboration, or authorship by Elliot Glazer is implied; the motivation is based only on publicly available work.

Abstract

A *blacklist system* assigns to each point x a set $F(x)$ of points that x refuses to pair with. A set is *safe* when no two of its members blacklist one another. This simple relation problem is a finite, quantitative cousin of Freiling’s axiom of symmetry and a natural bridge between extremal graph theory and choiceless mathematics.

For n points and blacklists of size at most k , put $q = 2k + 1$. We prove that every system has a safe set of size $\lceil n/q \rceil$, and that this is best possible for every n . Equivalently, a safe r -set is forced at the exact threshold

$$n = (r - 1)(2k + 1) + 1.$$

At the last failing value $n = (r - 1)q$, every extremal system is rigid: its conflict graph is a disjoint union of $r - 1$ copies of K_q , and each block is oriented as a regular tournament. This gives an exact enumeration of all labelled critical systems. We also determine the minimum number of safe pairs exactly, obtain the first two asymptotic orders for the number of safe r -sets for every fixed k, r , prove an exact formula for safe triples when $k = 1$, and establish a heterogeneous list-colouring theorem with local list size $2|F(x)| + 1$.

The infinite form exposes a foundational boundary. The classical compactness theorem of de Bruijn–Erdős yields a partition into q safe sets under a suitable choice principle. Over ZF we isolate the principle SAP_k and prove

$$\text{BPI} \implies \text{SAP}_k \implies \bigwedge_{2 \leq m \leq k+1} \mathbf{C}_m,$$

where BPI is the Boolean Prime Ideal Theorem and $\mathbf{C}_m$ is choice for families of m -element sets. We also prove SAP_k for every well-orderable underlying set in ZF. In the definable setting, regularity gives far more: a Borel relation with countable vertical sections on a perfect Polish space has a perfect safe set, while a measurable such relation on an atomless standard probability space admits an almost surely safe countably infinite i.i.d. sample. These results turn the proposed problem into a concrete research program about exact finite spectra, weak choice, Borel colourings, and generalized symmetry axioms.

[write] **Status and trust boundary, added 5 October 2026, batch 114.** This is an AI-assisted, unrefereed external manuscript (its status box says so), filed in ProveIt as the single-source research report **freiling-symmetry-blacklists** (provenance, the sources as the write read them, relation to the repository, the collected non-claims and the reading conventions are in Section 1.4). Its author line reads “Research report prepared for Vladimir Reshetnikov”. No statement of the report is formalized in Lean or Rocq; the formalization plan of Section 13 is a proposal. *What rests on what.* The partition of an arbitrary infinite system into $2k + 1$ safe sets is Theorem 3 of de Bruijn and Erdős [5], which the write read: the finite case of Corollary 3.3 and its low-degree-vertex induction are already there (note after Corollary 3.4). Turán’s theorem with its equality case is quoted without proof in Theorem 5.1; propositional compactness from BPI in Theorem 8.1; the Kuratowski–Ulam and Mycielski theorems in Theorem 9.1; Tonelli’s theorem in Theorem 9.3. Every other theorem, corollary, lemma and proposition is proved in the text; the intake and the write reread every proof and found no error. The exhaustive program `code/verification.py` checks the case $k = 1$ for $n \leq 7$, the count 40 by enumeration and the count 72,576 only by evaluating (6) (note in Section 11); it proves nothing beyond those cases. The source makes no historical-priority claim for its finite refinements “without a more exhaustive literature review” (delivery README), and the write made no such review either. The write adds the notes marked [write], Section 1.4 and Section 12.6 with three propositions proved there.

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1 Why this problem

1.1 The intersection of interests

Elliot Glazer’s dissertation studies interactions among fragments of the Axiom of Choice, forcing, measure theory, box-game paradoxes, and weak symmetry principles, explicitly including Freiling’s axiom of symmetry [1]. His paper on a choiceless box-game paradox asks how much of the familiar predictive phenomenon survives when choice is removed [2]. His topological Tennenbaum theorem is a separate example of extracting strong mathematical consequences from topological and Borel regularity assumptions [3].

The **ProveIt** repository already contains substantial work on measurable box games and infinite hat guessing [15, 16]. A useful new direction should therefore remain close to choice, symmetry, and definability without merely repeating that material. Sparse symmetry-avoidance does exactly this:

Finite face	exact extremal thresholds, equality cases, enumeration, independence polynomials, and algorithms;
Infinite face	compactness, weak choice, and the strength of a global finite partition;
Definable face	Borel category, atomless measure, perfect sets, and a regularity-based strengthening of Freiling’s conclusion.

1.2 Freiling symmetry in one line

Freiling’s axiom of symmetry concerns maps $f : \mathbb{R} \rightarrow [\mathbb{R}]^{\leq \aleph_0}$ and asserts the existence of x, y such that $y \notin f(x)$ and $x \notin f(y)$ [4]. In the language of this report, it asks for a safe pair when every point has a countable blacklist. The finite problem below replaces “countable” by a fixed bound k , asks for much larger safe sets and exact counts, and then returns to infinite sets to study global partitions and definable relations.

[write] **Freiling and Sierpiński, 5 October 2026.** The write confirmed the data of [4] (J. Symbolic Logic **51**, no. 1, March 1986, 190–200, doi:10.2307/2273955) and read its abstract and reference list on the publisher’s page, not its body. The abstract presents the axioms as based on “some intuition of Stuart Davidson and an old theorem of Sierpiński”; the reference list cites Sierpiński’s book [18], which the write did not read. The statement of the axiom above and its well-known equivalence with the negation of the continuum hypothesis in ZFC, usually credited to Sierpiński, are therefore not checked here against Freiling’s text or Sierpiński’s. This report proves only one direction of that equivalence, and only in the form of the remark after Corollary 9.2: under CH a well-ordering of $\mathbb{R}$ in type ω_1 gives countable blacklists with no safe pair (each of two distinct reals precedes the other or follows it). The converse, that Freiling’s axiom holds when CH fails, is neither stated nor used in this report.

1.3 The classical theorem that must be credited

De Bruijn and Erdős proved in 1951 that if $|F(x)| \leq k$ for all x , then the entire set is the union of $2k + 1$ safe sets [5]. Their paper explicitly adopts the Axiom of Choice and derives the infinite statement from graph-colouring compactness. The basic infinite partition is therefore classical. The point of this report is to refine the finite theory sharply, expose the exact extremizers, quantify the safe-set spectrum, and isolate the weak-choice and regularity questions that the classical formulation leaves hidden.

[write] **de Bruijn and Erdős, read 5 October 2026.** The write read the paper [5] in full (the scan of the reprint in the Eindhoven University of Technology repository). It confirms the paragraph above: “The Axiom of Choice is adopted throughout the paper”, and its Theorem 1 (a graph all of whose finite subgraphs

are k -colourable is k -colourable) is deduced from Rado’s selection theorem, with a proof by Tychonoff’s theorem credited to Rabson and A. Stone. Their Theorem 3 is Theorem 8.1 below: for $f(b) \subseteq S \setminus \{b\}$ with $|f(b)| \leq k$, S is the union of $2k + 1$ *independent* sets (their word for safe). Its proof for finite S is the argument of Lemma 3.1 and Corollary 3.3: the graph has at most $k|S|$ edges, hence a vertex joined to fewer than $2k + 1$ others, and induction on $|S|$; the infinite case follows from Theorem 1. Their Theorem 4 is the countable-colour statement of Question 12.3: if every $f(b)$ is finite, S is the union of countably many independent sets (apply Theorem 3 to each $S_k = \{b : |f(b)| = k\}$). The paper says that Theorems 1, 3 and 4 were announced in Erdős [17], which the write did not read. In the language of combinatorial set theory a map F with $x \notin F(x)$ is a *set mapping* and a safe set a *free set*; Hajnal’s free set theorem [19] (not read) belongs to that literature. The source does not use this vocabulary, and the write did not survey that literature; this is one more reason to keep the source’s priority caveat as it stands.

1.4 Provenance, sources and reading conventions

[write] **Added 5 October 2026, batch 114.** *Provenance.* The report is built from one manuscript, the archive `From_Finite_Blacklists_to_Freiling_Symmetry.zip` (wrapper directory of the same name, 6 files), which reached ProveIt’s docs/incoming in the arrival commit 2399df2bd (5 October 2026, “New research reports”), outside the session bundle of the intake, and was placed as batch 114 in 99053b5d1. The text is the delivered `From_Finite_Blacklists_to_Freiling_Symmetry.tex` (1,291 lines), shipped as `article.tex`; the delivered `verification.py` and `verification_output.txt` are shipped unchanged in `code/` and `data/`. The delivered 19-page PDF and `SHA256SUMS.txt` (four entries, all verified at placement) are not shipped; they and the delivery `README.txt` (staged at placement and replaced by the report `README`) are retrievable from the arrival commit. The package pins ProveIt revision `a763feee1f3703dc710d9b810b1de66168f1e266` (bibliography entry [15]); that commit is the batch-98A write commit of this repository and an ancestor of the commit `9ccf04eae` this write is based on. The status box calls the report “AI-assisted” and “unrefereed”; the author line reads “Research report prepared for Vladimir Reshetnikov”. The package contains no e-mail address, no “prepared for private review” line and no personal data. Single source: no merge choice was made. The write prefixed the 36 delivered labels with `fb1:` and updated every reference; it added the notes marked [write], this subsection, labels on eight delivered headings, one corollary, one conjecture and the thirteen questions (none had a label), four bibliography entries after the delivered ones, and Section 12.6 with Propositions 12.16–12.18. No statement, proof or number of the source was changed: the added propositions are the last numbered statements of their section and the added displays are unnumbered, so every delivered theorem and equation keeps its number.

The sources, as the write read them (5 October 2026).

- de Bruijn and Erdős [5]: read in full (note above). The reprint is headed “Reprinted from Proceedings, Series A, 54, No. 5 and Indag. Math., 13, No. 5, 1951” with the single pagination 371–373, and the paper was communicated at the meeting of 24 November 1951. The bibliography’s “Indagationes Mathematicae 13 (1951), 369–373” is not supported by that copy, which gives 371–373 for both.
- Freiling [4]: data, abstract and reference list read, body not read (note in Section 1.2).
- Läuchli [9] and Mycielski [12]: bibliographic data confirmed through Crossref; the papers were not read. Läuchli’s paper is in the delivered bibliography but cited nowhere in the text; see Question 12.2 in Section 12.6.
- Not read: Glazer’s dissertation and papers [1, 2, 3] (motivation only), Cowen [7, 8], Howard and Rubin [10], Kechris [13], and the uncited entries Turán [6], Jech [11] and Bollobás [14]. The standard facts the proofs take from them (propositional compactness from BPI, the Kuratowski–Ulam and Mycielski theorems, Turán’s theorem and its equality case) are used as the source states them.
- The OEIS: the counts of distinct conflict graphs in Section 11 are terms of A133686, labelled graphs whose components have at most one cycle [20], and $R_1, R_3, R_5 = 1, 2, 24$ are terms of A007079, labelled regular tournaments on $2k + 1$ nodes. The write identified both by searching the OEIS; the source names neither. OEIS data are licensed CC BY-SA 4.0; nothing was submitted.

What was checked at intake. The batch-114 dossier read the manuscript in full and checked the block construction, the Turán count $\binom{t}{2}q^2$, the arc-saturation step, formula (14), the component costs in the proof of Theorem 6.3 and the partition optimization, the clique-fibre choice function and the Lindenbaum recursion; no error. Independent programs confirmed the tables of $A_2(n, 1)$ and $A_3(n, 1)$ for $n \leq 12$ from the closed forms, the critical counts, $R_5 = 24$ by brute force, and I_{C_4}, I_{C_5} . The delivered verifier, run on a copy, printed its recorded output. The write reread every proof, found no error, reran the verifier on a copy (about 6 seconds, output identical up to line endings), and made the computation reported in Section 12.6.

Relation to the repository.

- *Formal status.* No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The plan of Section 13 is a proposal; at the write’s base commit no Lean or Rocq file of the repository mentions blacklists, free sets of set mappings, Freiling’s axiom, SAP_k or the Boolean prime ideal theorem.
- *Stale claims.* The source’s one statement about the repository, that it “already contains substantial work on measurable box games and infinite hat guessing”, is correct at the pin and at the write’s base commit: `measurable-box-games` (in this category, five Parts; title “Countable Information, Uncountably Many Boxes”) is unchanged between the two. Nothing needed correcting. The bibliography entry [16] names the repository’s owner as its author; that report’s README records how each of its Parts was produced.
- `measurable-box-games` is cited as background only: it shares the motivation from Glazer’s work and no theorem with this report.
- `bounded-width-power-set-compression` (same category and batch; *see also*): it too pairs an exact finite extremal theory with its choiceless infinite form over ZF (there, an infinite set admitting a bounded-width compression of its power set is Dedekind-infinite), and it too asks for the weakest standard choice principle that suffices, naming finite-choice schemes, the Boolean prime ideal theorem and compactness principles among the candidates, as Questions 12.1 and 12.2 do here. Neither source cites the other; no theorem is shared.
- Batch 114 also filed `noisy-parity-cubes`, `robust-neutral-choice` and `random-bits-arithmetic-cuts` in the same category; they share only the Glazer motivation and choice-built nonmeasurable objects, and no theorem.

[write] **What is not claimed, collected from the source.** The partition of an infinite system into $2k + 1$ safe sets is de Bruijn and Erdős’s (1951), “not . . . this report”, and their paper assumes the Axiom of Choice. “No exhaustive priority search has been made for the finite refinements proved here”, and the delivery README makes “no historical-priority claim . . . without a more exhaustive literature review”; the ledger of Section 14 must be read the same way (see the reading conventions below). No endorsement, collaboration or authorship by Elliot Glazer is implied; the motivation rests on his public work only. The computation “is verification, not proof”. The perfect safe set is “classical tools, derived here”: Kuratowski–Ulam and Mycielski, with the application to blacklists spelled out. The regularity theorems do not yield a partition of X into finitely many definable safe sets. The coefficientwise conjectures are “supported only by the proved coefficients and small exhaustive checks”, and the exact choice strength of SAP_k is open. The research program and the formalization plan are proposals.

[write] **Reading conventions.** In the ledger of Section 14, “no priority claim after an exhaustive search” means that no priority is claimed, as a claim would need an exhaustive search; it does not say that one was made (the status box says the opposite). “Proved here” in that ledger means proved in this text, not first proved here. The colour set $2k + 1$ in $c : X \rightarrow 2k + 1$ is the von Neumann ordinal $\{0, \dots, 2k\}$. “Safe” is de Bruijn and Erdős’s “independent” and the set-mapping literature’s “free”. The source reuses letters as follows; no symbol was renamed.

Symbol	Meaning here (where)	Not to be read as
$F, F(x)$	the blacklist system and the blacklist of x	the families $\mathcal{A}_e, \mathcal{A}$
$A_r(n, k), A$	the minimum number of safe r -sets (1); a safe set (Corollary 3.4) or a member of the family $\mathcal{A}$ (proof of Theorem 8.3)	each other
$\mathcal{A}_e, \mathcal{A}$	the r -sets containing the edge e (proof of Theorem 6.2); a family of m -element sets (Theorem 8.3)	each other
C_m, C_{fin}	choice for families of m -element sets, for families of nonempty finite sets (Section 12.6)	$C = R \cup R^{-1}$ (proof of Theorem 9.1); the cycle C_s ; countable choice
$q, Q(H)$	$q = 2k + 1$; the cost of a component H (proof of Theorem 6.3)	each other
$t, t(G)$	$t = r - 1$, the number of blocks (Section 5); the number of triangles of G	each other
R, R_q	a relation on X (Section 9); the number of labelled regular tournaments on q points	each other
H_n, H	the conjectured minimizer (Conjecture 12.5); a unicyclic component	each other
s, a	a block size (proof of Theorem 4.1); the remainder in $n = 3a + s$ and the order of a component (Theorem 6.3); a is also a point of A in the sandwich proof	
m	the number of edges in (14); the size of the sets in C_m	each other
c, c_n	a colouring or a colour; the correction (13)	
$b(x), L(x)$	$ F(x) $ and the list of x (Theorem 3.2)	
$e, e_F(S), e(G)$	a conflict edge; edge counts	
$i_r, I_G(z), \alpha$	the number of safe (independent) r -sets, the independence polynomial, the independence number	
$\text{SAP}_k, \text{SAP}_{\text{fin}}, \text{SAP}_{\text{fin}}^+, \text{BPI}$	the partition principle for k -bounded systems; its analogues for finite blacklists with countably many colours (Section 12.6); the Boolean prime ideal theorem	
P, μ	a perfect set; a probability measure	

2 The sparse symmetry-avoidance problem

Definition 2.1 (Blacklist system). Let X be a set and let $k \in \mathbb{N}$. A k -bounded blacklist system is a map

$$F : X \longrightarrow [X]^{\leq k}$$

with $x \notin F(x)$ for every $x \in X$. We draw an arc $x \rightarrow y$ when $y \in F(x)$.

Definition 2.2 (Safe set and conflict graph). Distinct $x, y \in X$ are *compatible* when

$$y \notin F(x) \quad \text{and} \quad x \notin F(y).$$

A subset $A \subseteq X$ is *safe* when every two distinct members are compatible. The *conflict graph* G_F is the undirected graph on X with

$$\{x, y\} \in E(G_F) \iff y \in F(x) \text{ or } x \in F(y).$$

Thus safe sets are exactly the independent sets of G_F .

For finite X , write $n = |X|$ and let $i_r(F) = i_r(G_F)$ denote the number of safe r -subsets. The central finite spectrum is

$$A_r(n, k) := \min_F i_r(F), \quad (1)$$

where the minimum ranges over all k -bounded systems on an n -element labelled set.

Problem 2.3 (Sparse symmetry-avoidance spectrum). Determine $A_r(n, k)$, the structure and number of its minimizers, and the corresponding infinite partition principle. In particular:

- (a) When is a safe r -set forced?
- (b) What happens at the last value of n for which it can fail?
- (c) How many safe r -sets are forced above threshold?
- (d) Which weak choice principle is exactly needed to partition every infinite system into $2k + 1$ safe sets?
- (e) What stronger conclusions follow from Borel or measurable regularity?

The finite and infinite questions use the same object but behave very differently. Finite systems admit a deterministic elimination algorithm. Infinite systems require a compactness step, and that step has genuine foundational content.

3 The graph translation and local sparsity

For $S \subseteq X$, let $G_F[S]$ be the induced conflict graph and let $e_F(S)$ be its number of edges.

Lemma 3.1 (Induced edge budget). *For every finite $S \subseteq X$,*

$$e_F(S) \leq \sum_{x \in S} |F(x) \cap S| \leq k|S|. \quad (2)$$

Proof. Every conflict edge in $G_F[S]$ is witnessed by at least one directed blacklist arc whose tail and head both lie in S . Counting such arcs gives the first inequality; the blacklist bound gives the second. $\square$

Consequently every nonempty finite induced subgraph has average degree at most $2k$ and hence a vertex of degree at most $2k$. A more useful local version allows the budget to vary from point to point.

Theorem 3.2 (Heterogeneous list theorem). *Let X be finite, put $b(x) = |F(x)|$, and assign a finite list of colours $L(x)$ to each $x \in X$. If*

$$|L(x)| \geq 2b(x) + 1 \quad (x \in X), \quad (3)$$

then there is a proper list-colouring of G_F ; equivalently, one may choose $c(x) \in L(x)$ so that every colour class is safe.

Proof. For a nonempty $S \subseteq X$, suppose every $x \in S$ had $d_{G_F[S]}(x) \geq 2b(x) + 1$. Then

$$2e_F(S) = \sum_{x \in S} d_{G_F[S]}(x) \geq 2 \sum_{x \in S} b(x) + |S|,$$

contradicting [Theorem 3.1](#). Hence some $x \in S$ has $d_{G_F[S]}(x) \leq 2b(x)$. Remove such a vertex repeatedly. Colour the vertices in reverse removal order. When x is restored, at most $2b(x)$ colours are forbidden by already coloured neighbours, while its list has at least $2b(x) + 1$ colours. $\square$

Corollary 3.3 (Finite partition bound). *Every finite k -bounded blacklist system is the union of*

$$q = 2k + 1$$

safe sets.

Corollary 3.4 (Weighted safe set). *If nonnegative weights $w(x)$ are assigned to a finite k -bounded system, then some safe set A satisfies*

$$\sum_{x \in A} w(x) \geq \frac{1}{2k + 1} \sum_{x \in X} w(x).$$

Proof. Apply [Theorem 3.3](#) and take a colour class of maximum total weight. $\square$

[write] **Credit, 5 October 2026.** Corollary 3.3, with the proof given here for constant lists (at most $k|S|$ edges, so a vertex of degree at most $2k$, then induction), is the finite case of Theorem 3 of de Bruijn and Erdős [5] and is proved there the same way (note in Section 1.3). Their paper has no list version: Theorem 3.2, with lists of size $2|F(x)| + 1$ varying from point to point, and Corollary 3.4 are not in it.

4 The exact finite threshold

The colouring bound immediately gives a safe set of size $\lceil n/q \rceil$. The next result shows that no improvement is possible, for any value of n .

Theorem 4.1 (Exact finite avoidance number). *For every $n \geq 1$ and $k \geq 0$,*

$$\min_F \alpha(G_F) = \left\lceil \frac{n}{2k + 1} \right\rceil. \quad (4)$$

Proof. The lower bound follows from [Theorem 3.3](#): one of the $q = 2k + 1$ colour classes has at least $\lceil n/q \rceil$ vertices.

For sharpness, partition the vertex set into blocks of size at most q , with exactly $\lceil n/q \rceil$ blocks. On a block of odd size $s = 2h + 1 \leq q$, use the cyclic regular tournament, so every vertex has outdegree $h \leq k$. On a block of even size $s = 2h \leq q - 1$, delete one vertex from a regular tournament on $s + 1$ vertices; every remaining outdegree is at most $h \leq k$. Let each point blacklist its out-neighbours in its block. The conflict graph is a disjoint union of cliques, one per block, so a safe set contains at most one point from each block. Its maximum size is therefore the number of blocks. $\square$

Corollary 4.2 (Sharp threshold for a safe r -set). *For $r \geq 2$, every k -bounded system on n points has a safe r -set if and only if*

$$n > (r - 1)(2k + 1). \quad (5)$$

Equivalently, the least forcing value is

$$n_{\min}(r, k) = (r - 1)(2k + 1) + 1.$$

This exact threshold is the first answer to [Theorem 2.3](#). Its equality case is much more rigid than the proof of the lower bound suggests.

5 Critical rigidity and exact enumeration

Put $q = 2k + 1$ and $t = r - 1$. The critical size is $n = tq$.

Theorem 5.1 (Classification at the last failing value). *Let F be a k -bounded blacklist system on $n = tq$ points, where $q = 2k + 1$ and $t \geq 1$. The following are equivalent.*

- (i) *F has no safe $(t + 1)$ -set.*
- (ii) *The conflict graph is a disjoint union of t copies of K_q .*
- (iii) *The vertices can be partitioned into t blocks of size q such that, on each block, the blacklist arcs form a regular tournament and there are no arcs between distinct blocks.*

Proof. The implications (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i) are immediate. Assume (i) and write $G = G_F$. The complement $\overline{G}$ contains no K_{t+1} , because such a clique would be a safe $(t + 1)$ -set. Turán's theorem, including its equality case, gives

$$e(\overline{G}) \leq \binom{t}{2} q^2.$$

Therefore

$$\begin{aligned} e(G) &\geq \binom{tq}{2} - \binom{t}{2} q^2 \\ &= \frac{tq(q-1)}{2} = ktq = kn. \end{aligned}$$

On the other hand, [Theorem 3.1](#) gives $e(G) \leq kn$. Equality holds throughout. Equality in Turán's theorem forces $\overline{G}$ to be the balanced complete t -partite graph, hence G is tK_q .

There are at most kn blacklist arcs, while the kn conflict edges each need at least one witnessing arc. Thus there is exactly one arc on every conflict edge, every point uses all k blacklist slots, and no pair is blacklisted in both directions. Each K_q is therefore a tournament with constant outdegree k , that is, a regular tournament. $\square$

The theorem explains why the odd number $2k + 1$ is not merely a consequence of averaging: at equality it is the unique block size supporting a complete conflict graph with exactly k outgoing arcs at every point.

Let R_q be the number of regular tournaments on a fixed labelled q -element vertex set. Regular tournaments exist because q is odd.

Corollary 5.2 (Number of labelled critical systems). *The number of labelled k -bounded systems on $n = tq$ points with no safe $(t + 1)$ -set is*

$$\boxed{\frac{(tq)!}{(q!)^t t!} R_q^t}. \tag{6}$$

Proof. Choose an unordered partition into t blocks of size q , then choose one of the R_q regular tournaments independently on each block. $\square$

For example, $R_3 = 2$ and $R_5 = 24$. Thus there are

$$\frac{6!}{(3!)^2 2!} 2^2 = 40$$

critical systems for $(k, r, n) = (1, 3, 6)$, and

$$\frac{10!}{(5!)^2 2!} 24^2 = 72,576$$

critical systems for $(k, r, n) = (2, 3, 10)$.

6 The safe-subset spectrum

The threshold asks only whether $A_r(n, k)$ vanishes. The more informative question is how rapidly it grows after the threshold.

6.1 Safe pairs: an exact formula

Theorem 6.1 (Exact pair spectrum). *For all $n \geq 1$ and $k \geq 0$,*

$$A_2(n, k) = \max \left\{ 0, \binom{n}{2} - kn \right\}. \quad (7)$$

Proof. There are at most kn conflict edges, so at least $\binom{n}{2} - kn$ pairs are safe. The count is nonnegative.

If $n \leq 2k + 1$, orient the complete graph with maximum outdegree at most k as in the proof of Theorem 4.1; there are no safe pairs. If $n > 2k + 1$, label the vertices by $\mathbb{Z}/n\mathbb{Z}$ and set

$$F(i) = \{i + 1, i + 2, \dots, i + k\} \pmod{n}.$$

No unordered pair receives two opposite arcs, so the conflict graph has exactly kn edges and the lower bound is attained. $\square$

6.2 All fixed orders: a sharp two-term asymptotic

We use the convention $\binom{a}{b} = 0$ when $b < 0$ or $b > a$.

Theorem 6.2 (Universal lower bound and asymptotic sharpness). *For every n, k, r ,*

$$A_r(n, k) \geq \max \left\{ 0, \binom{n}{r} - kn \binom{n-2}{r-2} \right\}. \quad (8)$$

If $n > 2k + 1$, then

$$\begin{aligned} A_r(n, k) &\leq \binom{n}{r} - kn \binom{n-2}{r-2} \\ &\quad + n \binom{2k}{2} \binom{n-3}{r-3} + \binom{kn}{2} \binom{n-4}{r-4}. \end{aligned} \quad (9)$$

Consequently, for fixed k and $r \geq 2$,

$$A_r(n, k) = \binom{n}{r} - kn \binom{n-2}{r-2} + O_{k,r}(n^{r-2}), \quad (10)$$

and hence

$$\frac{A_r(n, k)}{\binom{n}{r}} = 1 - \frac{kr(r-1)}{n-1} + O_{k,r}(n^{-2}). \quad (11)$$

Proof. For each conflict edge e , let $\mathcal{A}_e$ be the family of r -subsets containing e . Then $|\mathcal{A}_e| = \binom{n-2}{r-2}$. Since there are at most kn conflict edges, the union bound shows that at most $kn \binom{n-2}{r-2}$ subsets are unsafe, proving Equation (8).

For the upper bound on the minimum, use the circulant construction from Theorem 6.1. Its conflict graph is $2k$ -regular with kn edges. Bonferroni's inequality gives

$$\left| \bigcup_e \mathcal{A}_e \right| \geq \sum_e |\mathcal{A}_e| - \sum_{e < f} |\mathcal{A}_e \cap \mathcal{A}_f|.$$

There are exactly $n \binom{2k}{2}$ pairs of incident edges, each lying in $\binom{n-3}{r-3}$ common r -sets. The number of disjoint edge pairs is at most $\binom{kn}{2}$, and each lies in $\binom{n-4}{r-4}$ common r -sets. Subtracting the resulting lower bound on unsafe sets from $\binom{n}{r}$ proves Equation (9). The two error terms are $O_{k,r}(n^{r-2})$, and division by $\binom{n}{r}$ gives Equation (11). $\square$

The first correction has a transparent interpretation: to first order, each of the kn possible conflict edges destroys $\binom{n-2}{r-2}$ candidate safe sets. Overlaps among destroyed families enter one order later.

6.3 An exact third coefficient when $k = 1$

For $k = 1$, the conflict graphs are exactly the finite pseudoforests: every connected component contains at most one cycle. This permits an exact safe triple count.

Theorem 6.3 (Exact safe triples for one blacklist per point). *For $n < 3$, $A_3(n, 1) = 0$. For $n \geq 3$, write $n = 3a + s$ with $s \in \{0, 1, 2\}$. Then*

$$A_3(n, 1) = \binom{n}{3} - n(n-2) + c_n, \quad (12)$$

where

$$c_n = \begin{cases} 2a, & s = 0, \\ 2a + 2, & s = 1, \\ 2a + 3, & s = 2. \end{cases} \quad (13)$$

The bound is attained by a disjoint union of triangles when $s = 0$, by $(a-1)K_3 \sqcup C_4$ when $s = 1$, and by $(a-1)K_3 \sqcup C_5$ when $s = 2$.

Proof. A graph is the conflict graph of a 1-bounded system exactly when it is a pseudoforest. One direction follows from $e(S) \leq |S|$. Conversely, orient the unique cycle of each unicyclic component cyclically and every attached tree toward the cycle; orient a tree toward an arbitrary root. Every outdegree is at most one.

For any graph G with m edges, degrees $d(v)$, and $t(G)$ triangles, inclusion-exclusion on the edges contained in a triple gives

$$i_3(G) = \binom{n}{3} - m(n-2) + \sum_v \binom{d(v)}{2} - t(G). \quad (14)$$

Adding an edge without leaving the class of pseudoforests cannot increase the number of independent triples. Hence a minimizer may be assumed to have all components unicyclic, and therefore $m = n$.

For a unicyclic component H of order s , set

$$Q(H) = \sum_{v \in H} \binom{d(v)}{2} - t(H).$$

If $H = C_3$, then $Q(H) = 2$. If $s \geq 4$ and all degrees equal two, then $H = C_s$ and $Q(H) = s$. Otherwise the degree sum is $2s$ but the degrees are not all two, so integer convexity gives $\sum_v \binom{d(v)}{2} \geq s + 1$. A unicyclic component has at most one triangle, whence again $Q(H) \geq s$.

Thus a component of size three costs at least 2, while every other component of size $s \geq 4$ costs at least s . Minimizing the total cost over partitions of n into integers at least three gives exactly [Equation \(13\)](#): use as many 3's as possible, with one 4 or one 5 to absorb a nonzero remainder. Substitution into [Equation \(14\)](#) proves the result, and the displayed cycle unions attain equality. $\square$

[write] **A step made explicit, 5 October 2026.** The step “a minimizer may be assumed to have all components unicyclic” needs that a pseudoforest on $n \geq 3$ vertices to which no edge can be added without leaving the class has only unicyclic components. Suppose such a pseudoforest has a tree component T . If $|T| \geq 3$, two nonadjacent vertices of T exist, and joining them makes T unicyclic. If $|T| \leq 2$, then $n \geq 3$ gives another component U , and an edge from T to U yields a tree if U is a tree and a unicyclic component if U is unicyclic. Either way an edge can be added, a contradiction. Adding edges never increases i_3 , so starting from any minimizer and adding edges while possible reaches a minimizer with all components unicyclic, hence with $m = n$. The minimum over partitions is as stated: a part of size 3 costs $2 = 3 - 1$ and a part of size $s \geq 4$ costs at least s , so the total cost is at least n minus the number of parts of size 3, which is at most a when $s = 0$ and at most $a - 1$ otherwise (the remainder must be absorbed by a part of size at least 4); this gives $2a$, $2a + 2$, $2a + 3$.

The first values are:

n	1	2	3	4	5	6	7	8	9	10	11	12
$A_2(n, 1)$	0	0	0	2	5	9	14	20	27	35	44	54
$A_3(n, 1)$	0	0	0	0	0	0	6	15	27	48	75	108

7 Algorithms and certificates

The proof of [Theorem 3.2](#) is constructive.

Proposition 7.1 (Linear-time peeling after graph construction). *Given the blacklist lists explicitly, one can produce the elimination order and a safe $(2k + 1)$ -colouring in $O(n + \sum_x |F(x)| + e(G_F))$ time, using adjacency lists and a queue of vertices satisfying $d(x) \leq 2|F(x)|$.*

Algorithm. Construct the undirected conflict graph while deduplicating opposite arcs. Maintain current degrees and enqueue every currently eligible vertex. Removing a vertex can only decrease the degrees of its neighbours, so each vertex and edge is processed a constant number of times. Reverse the removal order and greedily choose an available colour (or an available list colour in the heterogeneous version). $\square$

This yields three useful certificates.

- A colouring certifies a safe set of size at least $\lceil n/(2k + 1) \rceil$ by its largest colour class.
- At the critical size $n = (r - 1)(2k + 1)$, failure of a safe r -set can be verified simply by checking the block decomposition and regular tournament conditions of [Theorem 5.1](#).

- A claimed extremizer can be checked in time linear in its explicit blacklist data, rather than by searching all r -subsets.

The optimization problem above threshold remains nontrivial: finding a maximum independent set is hard in broad sparse graph classes, even though the universal guarantee and its certificate are easy.

[write] **Scope of the third certificate, 5 October 2026.** The text proves a linear-time check for two kinds of extremizer only: the critical systems, through the block and regular-tournament conditions of Theorem 5.1 (second item), and the minimizers of $A_2(n, k)$, since $i_2(F) = \binom{n}{2} - e(G_F)$ and the proof of Theorem 6.1 shows that a system is a minimizer exactly when $e(G_F) = \min\{kn, \binom{n}{2}\}$. For a claimed minimizer of $A_r(n, k)$ with $r \geq 3$, or of $\alpha(G_F)$ above the critical size, the text gives no such check: verifying it needs the value $A_r(n, k)$, unknown in general (Question 12.4), or an upper bound on the independence number. The third item is therefore stated in Section 12.6 as an open question for those cases (item 9). The hardness sentence above is the source's, given without a reference.

8 The infinite partition and its choice content

8.1 The classical compactness extension

Theorem 8.1 (de Bruijn–Erdős, relation form). *Assume a graph-colouring compactness principle, for example the Boolean Prime Ideal Theorem. If $F : X \rightarrow [X]^{\leq k}$, then X is the union of $2k + 1$ safe sets.*

Proof. Every finite induced subsystem is $(2k + 1)$ -colourable by Theorem 3.3. Graph-colouring compactness extends the compatible finite constraints to a colouring of all of X . This is the relation problem proved by de Bruijn and Erdős [5]; the weaker hypothesis BPI suffices because it is equivalent to propositional compactness [7, 8, 10]. $\square$

The finite proof does not directly recurse through an infinite set. An infinite conflict graph may have no vertex of degree at most $2k$, because a point can be blacklisted by arbitrarily many others. Compactness is doing real work.

8.2 A precise ZF principle

Definition 8.2 (Sparse Avoidance Partition principle). For $k \geq 1$, SAP_k is the statement:

For every set X and every $F : X \rightarrow [X]^{\leq k}$ with $x \notin F(x)$, there exists $c : X \rightarrow 2k + 1$ such that every colour class is safe.

Theorem 8.3 (Choice sandwich). *Over ZF, for every $k \geq 1$,*

$$\text{BPI} \implies \text{SAP}_k \implies \bigwedge_{m=2}^{k+1} \text{C}_m, \quad (15)$$

where C_m denotes the axiom of choice for families of m -element sets.

Proof. The first implication is Theorem 8.1. For the second, fix $2 \leq m \leq k + 1$ and a family $\mathcal{A}$ of pairwise arbitrary m -element sets. Disjointify it by taking

$$X = \{(A, a) : A \in \mathcal{A}, a \in A\}.$$

For $x = (A, a)$, let

$$F(x) = \{(A, b) : b \in A \setminus \{a\}\}.$$

The blacklist size is $m - 1 \leq k$, and each fibre $\{A\} \times A$ induces a clique. A SAP_k colouring therefore assigns distinct colours to the m points in every fibre. Choose from A the unique element receiving the least colour used on that fibre. This defines a choice function for $\mathcal{A}$. $\square$

Since choice for pairs is not provable in ZF, neither is SAP_k for any $k \geq 1$. On the other hand, Equation (15) leaves a large gap: BPI is substantially stronger than the displayed bounded finite-choice principles. Locating SAP_k in that gap is one of the main open problems proposed here. **[write] 5 October 2026.** For $k = 1$ the lower end can be raised: SAP_1 also implies choice for triples (Proposition 12.17). The finite-blacklist analogue, de Bruijn and Erdős's countable-colour theorem, lies over ZF between BPI and choice for families of nonempty finite sets (Proposition 12.18). The comparison of BPI with the finite-choice principles in the preceding paragraph is stated without proof or reference (item 4 of Section 12.6).

8.3 Well-orderable sets require no additional choice

Theorem 8.4 (ZF theorem for well-orderable domains). *In ZF, if X is well-orderable and $F : X \rightarrow [X]^{\leq k}$, then X has a safe $(2k + 1)$ -colouring.*

Proof. Introduce propositional atoms $p_{x,c}$ for $x \in X$ and $c < 2k + 1$, and impose the finite clauses saying that each point receives exactly one colour and conflict neighbours receive different colours. Every finite set of clauses is satisfiable by Theorem 3.3.

Because X is well-orderable, the set of all formulas in this language is well-orderable. The usual Lindenbaum construction can therefore be carried out by transfinite recursion in ZF: at each formula φ , add either φ or $\neg\varphi$, choosing the first option when it preserves finite satisfiability and the second otherwise. At least one option preserves finite satisfiability, since failure of both would be witnessed by two finite contradictory subtheories. The resulting complete finitely satisfiable theory determines a truth assignment satisfying all original clauses, hence the desired colouring. $\square$

Thus any choiceless counterexample to SAP_k must live on a genuinely non-well-orderable set. This is exactly the kind of boundary between local finite mathematics and global choice that motivates Glazer's work.

9 Definable infinite relations: perfect and random safe sets

The unrestricted partition problem is choice-sensitive. Borel or measurable regularity produces a different and much stronger kind of conclusion: one can find a very large single safe set even when blacklists are merely countable.

9.1 A perfect safe set from Baire category

Theorem 9.1 (Perfect symmetry avoidance). *Let X be a perfect Polish space and let $R \subseteq X^2$ be a Borel irreflexive relation whose vertical sections*

$$R_x = \{y : (x, y) \in R\}$$

are countable. Then there is a nonempty perfect set $P \subseteq X$ such that for all distinct $x, y \in P$,

$$(x, y) \notin R \quad \text{and} \quad (y, x) \notin R.$$

In blacklist language, P is a perfect safe set.

Proof. Every countable subset of a perfect Polish space is meagre. By the Kuratowski–Ulam theorem, the Borel set R is meagre in X^2 , because all of its vertical sections are meagre. Coordinate exchange is a homeomorphism, so R^{-1} is meagre as well. The symmetric conflict relation $C = R \cup R^{-1}$ is therefore meagre. Mycielski’s theorem supplies a perfect set P whose distinct pairs avoid C [12, 13]. $\square$

Corollary 9.2 (Definable strengthening of Freiling symmetry). *A Borel countable-valued blacklist relation on $\mathbb{R}$ not only has a safe pair; it has a perfect, hence continuum-sized, safe set.*

This theorem cleanly separates regular assignments from the classical counterexample under CH. If the reals are well-ordered in type ω_1 , letting each real blacklist all of its predecessors gives countable blacklists and destroys every safe pair. Such a relation cannot be Borel.

9.2 An almost surely safe countable sample

Theorem 9.3 (Random countable avoidance). *Let (X, μ) be an atomless standard probability space and let $R \subseteq X^2$ be product-measurable with every vertical section R_x countable. If $(X_n)_{n \in \mathbb{N}}$ are independent samples with law μ , then with probability one:*

- (i) *the points X_n are all distinct; and*
- (ii) *for every $i \neq j$, neither (X_i, X_j) nor (X_j, X_i) belongs to R .*

Thus the random range $\{X_n : n \in \mathbb{N}\}$ is almost surely a countably infinite safe set.

Proof. Atomlessness makes every countable set null. Tonelli’s theorem gives

$$(\mu \times \mu)(R) = \int_X \mu(R_x) d\mu(x) = 0.$$

The inverse relation is also null, by coordinate symmetry, and the diagonal is null. For each ordered pair $i \neq j$, the event that (X_i, X_j) lies in $R \cup R^{-1}$ has probability zero; for each $i < j$, the event $X_i = X_j$ has probability zero. There are only countably many such events, so their union is null. $\square$

For every fixed r , an equivalent finite-dimensional statement is that μ^r -almost every r -tuple of distinct points is safe. The countable version makes the regularity amplification more vivid: measurable countable blacklists are almost surely avoided simultaneously by an entire random sequence.

9.3 What these regularity theorems do not say

Neither Theorems 9.1 and 9.3 automatically yields a partition of all of X into finitely many definable safe sets. Perfect-set extraction, almost-sure sampling, ordinary finite colouring, Borel colouring, and measurable colouring are different problems. Their separation is a major source of further questions in Section 12.

10 Finite versus infinite: the phase diagram

The same blacklist relation has four sharply different regimes.

Regime	Hypothesis	Best conclusion established here
Finite, arbitrary	$ X = n, F(x) \leq k$	Exact safe number $\lceil n/(2k+1) \rceil$; exact threshold; critical rigidity; quantitative spectrum.
Infinite, arbitrary	no regularity	A $(2k+1)$ -partition under BPI; in ZF for well-orderable X ; exact weak choice strength open.
Topological	perfect Polish X , Borel countable sections	A perfect safe set.
Measure-theoretic	atomless standard probability, measurable countable sections	Almost every i.i.d. countable sample is safe.

This is the central conceptual answer to the design request. The finite form is exact and algorithmic. The unrestricted infinite form is governed by compactness and weak choice. The definable infinite form is dramatically more regular than either: category and measure amplify a pairwise symmetry intuition into perfect or countably infinite safe sets.

11 Computational verification

The accompanying script `verification.py` performs exhaustive checks for all 1-bounded systems on at most seven labelled vertices. A vertex has n choices (no blacklist target, or one of the other $n-1$ vertices), so the raw search contains n^n systems. Systems are deduplicated by conflict graph before independent-set counting.

The executed output is:

```
Exhaustive minima for k=1
n  distinct conflict graphs  A_2  A_3  alpha_min
1      1          1      0      0          1
2      2          2      0      0          1
3      8          8      0      0          1
4     57         57      2      0          2
5    608        608      5      0          2
6   8524       8524      9      0          2
7  145800     145800     14      6          3
```

```
Critical systems at (k,r,n)=(1,3,6): 40
Labeled regular tournaments: R_3=2, R_5=24
Critical systems at (k,r,n)=(2,3,10): 72576
```

All checks passed.

The computation is verification, not proof. The proofs in [Theorems 4.1, 5.1 and 6.1 to 6.3](#) are independent of the search.

[write] **What the program computes, 5 October 2026.** The shipped code/`verification.py` (standard library only) does four things. (1) For each $n \leq 7$ it runs through all n^n one-bounded systems, collects their distinct conflict graphs (the second column, which is A133686 of the OEIS, labelled graphs whose components have at most one cycle, as the proof of [Theorem 6.3](#) predicts), and computes the coefficientwise

minimum of their independence polynomials; it asserts that the coefficients of z^2 and z^3 equal (7) and (12) with $k = 1$, and that the largest r with a positive minimum is $\lceil n/3 \rceil$. The column headed `alpha_min` prints that theorem value after the assertion. (2) It counts the systems on six points with no safe triple among all $6^6 = 46,656$ one-bounded systems: 40, by enumeration. (3) It finds $R_3 = 2$ and $R_5 = 24$ by running through all orientations of K_3 and K_5 . (4) It obtains 72,576 by evaluating (6) with $R_5 = 24$; no two-bounded system is enumerated, so this count rests on Corollary 5.2. The program writes no file. The write reran it on a copy (Windows, Python 3.14.4): about 6 seconds, output identical to `data/verification_output.txt` up to line endings. For $n \leq 7$ every coefficient of I_{H_n} beyond z^3 vanishes, so these checks do cover every coefficient of Conjecture 12.5 there, as the text after it says; Proposition 12.16 turns this into a proof for $n \leq 9$.

12 A proposed research program

12.1 The choice-theoretic core

Research question 12.1 (Exact strength of SAP_1). Is SAP_1 equivalent over ZF to choice for pairs, to a known permutation or multiple-choice principle, or to a strictly stronger principle?

The directed structure for $k = 1$ is a partial functional graph. Its components may have cycles, one-way tails, or highly branching inverse trees. Colouring all components simultaneously appears to require choosing phases or markers, but much less than arbitrary propositional compactness.

Research question 12.2 (The hierarchy SAP_k). Do the principles SAP_k form a strict hierarchy? Does some finite k reach BPI? What are their implications among the standard principles in Howard–Rubin’s catalogue [10]?

Research question 12.3 (Choice-free countable-colour theorem). De Bruijn and Erdős also proved that finite, but not uniformly bounded, blacklists admit a countable safe partition under their ambient choice assumptions. What is the exact ZF strength of that statement?

12.2 The finite spectrum

Research question 12.4 (Exact $A_r(n, k)$). Determine $A_r(n, k)$ beyond pairs and the case $(r, k) = (3, 1)$. The asymptotic in Theorem 6.2 fixes the first two orders but leaves an $O(n^{r-2})$ optimization problem involving overlap patterns among conflict edges.

For a graph G , let

$$I_G(z) = \sum_{r \geq 0} i_r(G) z^r$$

be its independence polynomial.

Conjecture 12.5 (Coefficientwise pseudoforest minimizer). *For $n \geq 3$, among all n -vertex pseudoforests, the independence polynomial is coefficientwise minimized by*

$$H_n = \begin{cases} aK_3, & n = 3a, \\ (a-1)K_3 \sqcup C_4, & n = 3a+1, \\ (a-1)K_3 \sqcup C_5, & n = 3a+2. \end{cases}$$

Equivalently,

$$I_G(z) \succeq \begin{cases} (1+3z)^a, \\ (1+3z)^{a-1}(1+4z+2z^2), \\ (1+3z)^{a-1}(1+5z+5z^2), \end{cases}$$

according to $n \bmod 3$.

The theorem on safe triples proves one coefficient of this conjecture, and the exhaustive search verifies all coefficients for $n \leq 7$.

[write] 5 October 2026. Theorem 6.1 with $k = 1$ proves the coefficient of z^2 as well. Proposition 12.16 proves both conjectures in their smallest cases (this one for $n \leq 9$), and an exact computation of the write confirms this one for $n \leq 18$ (item 1 of Section 12.6).

Conjecture 12.6 (Critical block minimization). *When $q = 2k + 1$ divides n , the disjoint union $(n/q)K_q$ minimizes the independence polynomial coefficientwise among conflict graphs of k -bounded systems.*

The divisibility hypothesis matters. For example, with $k = 1$ and $n = 4$, C_4 has fewer independent pairs than $K_3 \sqcup K_1$.

Research question 12.7 (Stability and supersaturation). If a critical or near-critical system has very few safe r -sets, must it be close, in edit distance, to a union of regular-tournament blocks? Find sharp supersaturation inequalities above $n = (r - 1)(2k + 1)$.

Research question 12.8 (Enumeration). Enumerate all minimizers of $A_r(n, k)$ and obtain asymptotics for their number. At the exact threshold the answer is Equation (6); beyond it, one expects a mixture of tournament counts and sparse stability theory.

12.3 Definability and regularity

Research question 12.9 (Borel partition number). For a Borel k -bounded blacklist relation on a standard Borel space, is a Borel safe partition into finitely many pieces always possible? If not, what is the optimal Borel chromatic bound, and how does it depend on additional bounds on incoming sections?

Research question 12.10 (Measurable positive safe sets). Under the hypotheses of Theorem 9.3, when must there exist a measurable safe set of positive measure? What is the best universal measure bound under measure-preserving, bounded-indegree, or graphing assumptions?

Research question 12.11 (Regularity strength in ZF). Which fragments of dependent choice or Baire-category regularity suffice for Theorems 9.1 and 9.3? Can their exact foundational strengths be separated from SAP_k ?

12.4 Cardinal, probabilistic, and higher-arity extensions

Research question 12.12 (Cardinal blacklists). For infinite cardinals κ , study assignments $F : X \rightarrow [X]^{\leq \kappa}$. Determine the least number of safe classes, the largest forced safe cardinality, and the relation to generalized Freiling principles and cardinal arithmetic.

Research question 12.13 (Random blacklist systems). In a random k -out directed model, determine the typical independence number, safe-set polynomial, fluctuations, and the gap between typical and worst-case behaviour. The worst-case value is $n/(2k + 1)$, while random sparse graphs should have substantially larger safe sets.

Research question 12.14 (Hypergraph avoidance). Allow each point to forbid configurations of arity $d > 2$. Find the finite threshold, compactness strength, and measurable perfect-set analogues for safe subsets avoiding all directed forbidden patterns.

Research question 12.15 (Game and online forms). Study a game in which an adversary reveals blacklists online and a player must colour immediately, or chooses a large safe set under limited queries. Determine competitive ratios and whether randomization changes the sharp constant $2k + 1$.

12.5 Sequence and formalization projects

The arrays $A_r(n, k)$, the number of minimizers, and the labelled critical counts Equation (6) are natural candidates for an OEIS-oriented subproject once enough terms and independent literature checks are available. The regular tournament factor R_{2k+1} links the problem to existing tournament enumeration, while the new two-parameter structure records the finite–infinite symmetry problem rather than tournament counts alone.

[write] **5 October 2026.** R_{2k+1} is A007079 of the OEIS [20] (labelled regular tournaments on $2k + 1$ nodes; $R_7 = 2640$). Nothing has been submitted to the OEIS. The computation reported in Section 12.6 (item 1) gives $A_r(n, 1)$ for every r and every $n \leq 18$.

12.6 Further questions and research

[write] **Added 5 October 2026, batch 114.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. The two conjectures and the thirteen research questions above are the source’s own proposals; all are open, three are re-scoped below by Propositions 12.16–12.18, which the write proves, and one by an exact computation. Four sentences of the body that assert more than is proved are attached to the items they belong to: the third certificate of Section 7 (a new item), the comparison of BPI with the finite-choice principles after Theorem 8.3, the sketch after Question 12.1, and the expectations in Questions 12.8 and 12.13. The intake and the write found no false statement in the source.

1. **Coefficientwise pseudoforest minimizer** (Conjecture 12.5). Proved: the coefficients of $z^0, \dots, z^3$ (Theorems 6.1 and 6.3), and with them the whole conjecture for $n \leq 9$ (Proposition 12.16). *Observation of the write* (exact computation, 5 October 2026; program not shipped): the conjecture holds for every $n \leq 18$, so that $A_r(n, 1) = [z^r]I_{H_n}$ for all r and $n \leq 18$. Method: by the reduction in the proof of Theorem 6.3 (every pseudoforest on $n \geq 3$ vertices is a spanning subgraph of one whose components are all unicyclic, and adding edges lowers every coefficient), it suffices to range over all-unicyclic graphs; a unicyclic component is a cycle with a rooted tree at each cycle vertex, its independence polynomial is the trace of a product of 2×2 transfer matrices with entries the polynomials of the rooted trees with the root in or out, and every quantity is a polynomial with nonnegative coefficients in those entries, so only coefficientwise minimal objects need to be kept at each stage. The final set of minimal polynomials is $\{I_{H_n}\}$ for every $3 \leq n \leq 18$. Cross-checks: without that pruning, all distinct independence polynomials of all-unicyclic graphs were enumerated for $n \leq 11$ (575 for $n = 11$), and the coefficientwise minima for $n \leq 7$ agree with a brute-force run of the delivered program over all n^n systems. Missing: a proof for $n \geq 10$; the first coefficient not covered by Proposition 12.16 is that of z^4 at $n = 10$, where $[z^4]I_{H_{10}} = 18$.
2. **Critical block minimization** (Conjecture 12.6). Proved: the coefficients of z^0, z^1, z^2 and of every z^r with $r > n/q$, hence the whole conjecture for $n \in \{q, 2q\}$ (Proposition 12.16); for $k = 1$ it is the case $3 \mid n$ of item 1, so it holds for $n \leq 18$. Missing: every case with $k \geq 2$ and $n \geq 3q$, beginning with the coefficient of z^3 at $(k, n) = (2, 15)$, which is 125 for $3K_5$.
3. **Exact strength of SAP_1** (Question 12.1). Proved: $SAP_1 \Rightarrow C_2$ (Theorem 8.3) and, by Proposition 12.17, $SAP_1 \Rightarrow C_3$. So SAP_1 is equivalent to choice for pairs only if C_2 implies C_3 over ZF; the write did not consult the literature on implications among the principles C_m . The source’s sketch (colouring the components of a partial functional graph “appears to require choosing phases or markers, but much less than arbitrary propositional compactness”) is a heuristic. Missing: any upper bound on SAP_1 below BPI other than well-orderability of the domain (Theorem 8.4).

4. **The hierarchy SAP_k** (Question 12.2). Proved: $\text{BPI} \Rightarrow \text{SAP}_k \Rightarrow \bigwedge_{2 \leq m \leq k+1} \text{C}_m \wedge \text{C}_3$, and BPI implies a principle $\text{SAP}_{\text{fin}}^+$ that implies every SAP_k (Proposition 12.18). The sentence after Theorem 8.3, “BPI is substantially stronger than the displayed bounded finite-choice principles”, is given without proof or reference; read as “ $\bigwedge_{2 \leq m \leq k+1} \text{C}_m$ does not imply BPI over ZF”, it was not checked by the write, and the open problem of locating SAP_k in the gap presupposes it. Läuchli’s paper [9], in the source’s bibliography but cited nowhere in its text, bears on the last clause: by its abstract (seen by the write only in a search-engine excerpt of the publisher’s page) it assigns to every Boolean algebra a graph whose finite subgraphs are 3-colourable and whose 3-colourability yields a prime ideal. Such graphs need not be conflict graphs of k -bounded systems, and the write did not examine whether the construction can be adapted. Missing: whether some SAP_k implies SAP_{k+1} or BPI, and their places in the Howard–Rubin catalogue.
5. **The countable-colour theorem** (Question 12.3). The statement is Theorem 4 of de Bruijn and Erdős (read; note in Section 1.3), whose proof applies their Theorem 3 to each class S_k under the Axiom of Choice. Re-scoped by Proposition 12.18: over ZF it lies between BPI and choice for families of nonempty finite sets. Missing: its exact position, and whether it implies any SAP_k .
6. **Exact $A_r(n, k)$** (Question 12.4). Known exactly: $r = 2$ (Theorem 6.1), $(r, k) = (3, 1)$ (Theorem 6.3), the zero range (Corollary 4.2), and, from item 1, every r for $k = 1, n \leq 18$ (computation). Missing: everything else, in particular the $O(n^{r-2})$ term of (10).
7. **Stability and supersaturation** (Question 12.7). Proved: rigidity at the critical size (Theorem 5.1). Missing: any stability or supersaturation statement.
8. **Enumeration of minimizers** (Question 12.8). Proved: the critical count (6). The expectation beyond threshold (“a mixture of tournament counts and sparse stability theory”) is a heuristic. Missing: any count beyond threshold.
9. **Linear-time certificates for extremizers** (the third item of the list in Section 7). Proved: linear-time checks for critical systems and for minimizers of $A_2(n, k)$ (note there). Missing: a check for claimed minimizers of $A_r(n, k)$, $r \geq 3$, or of $\alpha(G_F)$ above the critical size, that does not search the r -subsets.
10. **Borel partition number** (Question 12.9). Proved: perfect safe sets for countable sections (Theorem 9.1), which is not a partition (the source’s own caveat). Missing: everything asked.
11. **Measurable positive safe sets** (Question 12.10). Proved: almost surely safe countable samples (Theorem 9.3). Missing: everything asked.
12. **Regularity strength in ZF** (Question 12.11). The proofs of Theorems 9.1 and 9.3 are written in ZFC. Missing: the fragments of choice they need, and their separation from SAP_k .
13. **Cardinal blacklists** (Question 12.12). Proved here: nothing; the classical free-set literature for set mappings of infinite order (note in Section 1.3) was not surveyed by the source or the write. Missing: everything asked.
14. **Random blacklist systems** (Question 12.13). The worst case is $\lceil n/(2k+1) \rceil$ (Theorem 4.1; the question prints $n/(2k+1)$). That random sparse systems “should have substantially larger safe sets” is a heuristic of the source, not proved. Missing: everything asked.
15. **Hypergraph avoidance** (Question 12.14). Missing: everything asked.

16. Game and online forms (Question 12.15). Missing: everything asked.

17. Formalization and sequences. The plan M1–M7 of Section 13 and the OEIS subproject above are proposals; nothing is formalized or submitted.

Proposition 12.16 ([write] Small cases, 5 October 2026). (a) *Let $n \geq 3$ and let G be an n -vertex pseudoforest. Then $i_r(G) \geq i_r(H_n)$ for $r \leq 3$ and for every $r > \deg I_{H_n}$. Here the degree of I_{H_n} is a if $n = 3a$, and $a + 1$ if $n = 3a + 1$ or $n = 3a + 2$. Consequently Conjecture 12.5 holds for $3 \leq n \leq 9$.*

(b) *Let $k \geq 0$, $q = 2k + 1$, $n = tq$ with $t \geq 1$, and let G be the conflict graph of a k -bounded system on n points. Then $i_r(G) \geq i_r(tK_q)$ for $r \leq 2$ and for every $r \geq t + 1$. Consequently Conjecture 12.6 holds for $n \in \{q, 2q\}$.*

Proof. (a) By the first paragraph of the proof of Theorem 6.3, the n -vertex pseudoforests are exactly the conflict graphs of one-bounded systems on n points. For every graph, $i_0 = 1$ and $i_1 = n$. By Lemma 3.1 with $k = 1$ and $S = X$, G has at most n edges, so $i_2(G) = \binom{n}{2} - e(G) \geq \binom{n}{2} - n$; each component K_3, C_4, C_5 of H_n has as many edges as vertices, so H_n has exactly n edges and $i_2(H_n) = \binom{n}{2} - n$. By Theorem 6.3, $i_3(G) \geq A_3(n, 1)$, and the graphs that the theorem names as attaining $A_3(n, 1)$ are aK_3 , $(a - 1)K_3 \sqcup C_4$ and $(a - 1)K_3 \sqcup C_5$, that is, H_n ; so $i_3(G) \geq i_3(H_n)$. The independence polynomial of a disjoint union is the product of those of the components, so I_{H_n} is $(1 + 3z)^a$, $(1 + 3z)^{a-1}(1 + 4z + 2z^2)$ or $(1 + 3z)^{a-1}(1 + 5z + 5z^2)$, of degree a , $a + 1$ and $a + 1$ (all coefficients of the factors are positive), and $i_r(H_n) = 0 \leq i_r(G)$ for r beyond the degree. For $n = 3, \dots, 9$ the degrees are 1, 2, 2, 2, 3, 3, 3, at most 3, so every r falls under one of the two cases.

(b) For every graph, $i_0 = 1$ and $i_1 = n$. By Lemma 3.1, $e(G) \leq kn$, so $i_2(G) \geq \binom{n}{2} - kn$; and tK_q has $t\binom{q}{2} = tqk = kn$ edges, so $i_2(tK_q) = \binom{n}{2} - kn$. An independent set of tK_q has at most one vertex in each of the t cliques, so $i_r(tK_q) = 0 \leq i_r(G)$ for $r \geq t + 1$. For $t \leq 2$ every r satisfies $r \leq 2$ or $r \geq t + 1$. $\square$

Proposition 12.17 ([write] SAP_1 gives choice for triples, 5 October 2026). *Over ZF, SAP_k implies C_3 for every $k \geq 1$. In particular $\text{SAP}_1 \Rightarrow \text{C}_2 \wedge \text{C}_3$. (For $k \geq 2$ this is already part of Theorem 8.3; the new case is $k = 1$.)*

Proof. Let $\mathcal{A}$ be a family of 3-element sets. For $A \in \mathcal{A}$ let $\Sigma(A)$ be the set of permutations of A of order 3; it has exactly two elements, σ and σ^{-1} , and is defined from A without any choice. Put

$$X = \{(A, \sigma, a) : A \in \mathcal{A}, \sigma \in \Sigma(A), a \in A\}, \quad F(A, \sigma, a) = \{(A, \sigma, \sigma(a))\}.$$

Since σ has no fixed point, $(A, \sigma, a) \notin F(A, \sigma, a)$ and $|F(x)| = 1 \leq k$. On each fibre $\{A\} \times \{\sigma\} \times A$ the arcs $a \rightarrow \sigma(a) \rightarrow \sigma^2(a) \rightarrow a$ join every two of its three points, so every fibre is a clique of the conflict graph. SAP_k gives $c : X \rightarrow 2k + 1$ with safe colour classes; c is injective on each fibre. Let $s(A, \sigma)$ be the point $a \in A$ for which $c(A, \sigma, a)$ is least. Define $g(A) = s(A, \sigma)$ if $s(A, \sigma) = s(A, \sigma^{-1})$, and otherwise let $g(A)$ be the unique element of $A \setminus \{s(A, \sigma), s(A, \sigma^{-1})\}$. The definition is symmetric in σ and σ^{-1} , so $g(A)$ depends only on A and c , and $g(A) \in A$. Thus g is a choice function on $\mathcal{A}$, defined in ZF from the single witness c . The second sentence follows with Theorem 8.3 for $m = 2$. $\square$

Let SAP_{fin} be the statement that for every set X and every $F : X \rightarrow [X]^{<\omega}$ with $x \notin F(x)$ there is $c : X \rightarrow \omega$ whose colour classes are safe (the conclusion of Theorem 4 of [5]), $\text{SAP}_{\text{fin}}^+$ the same statement with the extra requirement $c(x) \leq 2|F(x)|$ for every x , and C_{fin} the axiom of choice for families of nonempty finite sets.

Proposition 12.18 ([write] Countable colours over ZF, 5 October 2026). *Over ZF,*

$$\text{BPI} \implies \text{SAP}_{\text{fin}}^+ \implies \text{SAP}_{\text{fin}} \implies \text{C}_{\text{fin}}, \quad \text{and} \quad \text{SAP}_{\text{fin}}^+ \implies \text{SAP}_k \quad \text{for every } k \geq 1.$$

Proof. $\text{BPI} \implies \text{SAP}_{\text{fin}}^+$. Let $F : X \rightarrow [X]^{<\omega}$, $b(x) = |F(x)|$ and $L(x) = \{0, 1, \dots, 2b(x)\}$. Take propositional variables $p_{x,j}$ for $x \in X$ and $j \in L(x)$, and the clauses $\bigvee_{j \in L(x)} p_{x,j}$ (a finite disjunction), $\neg(p_{x,i} \wedge p_{x,j})$ for $i \neq j$ in $L(x)$, and $\neg(p_{x,j} \wedge p_{y,j})$ for every conflict edge $\{x, y\}$ and $j \in L(x) \cap L(y)$. A finite set of these clauses mentions a finite set S of points. On S consider the system $F_S(x) = F(x) \cap S$; its conflict graph is the induced graph $G_F[S]$, and $|L(x)| = 2b(x) + 1 \geq 2|F_S(x)| + 1$. Theorem 3.2 gives a proper list colouring of $G_F[S]$ from the lists $L(x)$, and setting $p_{x,j}$ true exactly when x has colour j satisfies the finite set. BPI is equivalent over ZF to the compactness theorem of propositional logic (the fact used in Theorem 8.1), so all clauses are satisfied by one valuation; let $c(x)$ be the unique $j \in L(x)$ with $p_{x,j}$ true. Then $c(x) \leq 2b(x)$, and two conflicting points get different colours, so every colour class is safe.

$\text{SAP}_{\text{fin}}^+ \implies \text{SAP}_{\text{fin}}$ is trivial, and $\text{SAP}_{\text{fin}}^+ \implies \text{SAP}_k$: for a k -bounded system, $c(x) \leq 2k$, so c maps into $2k + 1$.

$\text{SAP}_{\text{fin}} \implies \text{C}_{\text{fin}}$. This is the clique-fibre argument of Theorem 8.3 without the size bound. Let $\mathcal{A}$ be a family of nonempty finite sets, $X = \{(A, a) : A \in \mathcal{A}, a \in A\}$ and $F(A, a) = \{(A, b) : b \in A \setminus \{a\}\}$, a finite set not containing (A, a) . Each fibre $\{A\} \times A$ is a clique of the conflict graph, so a colouring $c : X \rightarrow \omega$ with safe classes is injective on it, and choosing from A the element of least colour defines a choice function on $\mathcal{A}$ in ZF. $\square$

13 Formalization plan for ProveIt

A compact Lean development can be organized into the following modules.

- M1. **Blacklist and conflict graph.** Finite types, the directed relation $y \in F(x)$, symmetrization, and equivalence between safe sets and independent sets.
- M2. **Edge budget.** Map each conflict edge to a witnessing arc, prove $e(S) \leq \sum_{x \in S} |F(x) \cap S|$, and derive a low-degree vertex.
- M3. **Peeling and list colouring.** Formalize the variable-budget elimination order and greedy reverse colouring.
- M4. **Sharp constructions.** Cyclic tournaments, near-regular even orientations, block sums, and the exact independence-number theorem.
- M5. **Critical equality.** Import or prove the finite Turán equality case, then derive the regular-tournament classification and counting formula.
- M6. **Spectrum.** Double counting for pairs, Bonferroni bounds, and the pseudoforest triple formula.
- M7. **Foundations interface.** State the infinite colouring as an axiom parameterized by a compactness principle; keep the finite theorem free of classical-choice dependencies.

The finite core should require only ordinary finite combinatorics from Mathlib. The descriptive-set-theoretic perfect-set theorem is a separate, substantially larger formalization project.

14 Assumption, novelty, and claim ledger

Claim or ingredient	Status	Notes
$2k + 1$ -colouring of arbitrary infinite systems	Classical	De Bruijn–Erdős (1951), via compactness; their paper adopts AC.
Finite list theorem and exact safe number	Proved here	Elementary edge-budget and greedy arguments; no priority claim after an exhaustive search.
Critical regular-tournament classification	Proved here	Uses Turán’s equality theorem and saturation of the arc budget.
Exact critical enumeration	Proved here	Immediate from the classification and labelled regular tournament count.
Exact pair spectrum	Proved here	Edge count plus an explicit circulant construction.
Fixed- r two-term asymptotic	Proved here	Union bound and second Bonferroni inequality.
Exact $A_3(n, 1)$	Proved here	Pseudoforest reduction and a component cost inequality.
Choice sandwich	Proved here	BPI upper bound by compactness; bounded finite-choice lower bounds by clique fibres.
Perfect safe set	Classical tools, derived here	Kuratowski–Ulam plus Mycielski; the application to blacklist symmetry is spelled out.
Random countable safe set	Proved here from standard measure theory	Tonelli plus a countable union of null events.
Coefficientwise minimization conjectures	Open	Supported only by the proved coefficients and small exhaustive checks.
Exact choice strength of SAP_k	Open	The central foundational question proposed by the report.

15 Conclusion

Sparse symmetry-avoidance is deliberately simple to state:

Each point names at most k points it refuses to pair with. How large a mutually accepting set must exist, and how much choice is needed to organize all points into finitely many such sets?

The finite answer begins with a complete theorem: the exact denominator is $2k + 1$, the safe- r threshold is $(r - 1)(2k + 1) + 1$, and the final obstructions are precisely blocks of regular tournaments. The quantitative theory then opens into safe-set spectra, pseudoforest formulas, stability, and enumeration.

The infinite answer is more revealing. Local finite colourability alone does not specify which compactness principle may be used to assemble a global colouring. The principle SAP_k is strong enough to imply genuine finite-choice axioms, weak enough that its exact position below BPI is unclear, and automatically true on well-orderable sets. Meanwhile Borel and measurable regularity

bypass the partition question and produce perfect or random countably infinite safe sets. This combination of exact finite combinatorics, weak choice, Freiling symmetry, and definability makes the problem a natural addition to the ProveIt research program and a plausible point of contact with Glazer’s published interests.

A A self-contained compactness lemma for well-orderable languages

For completeness, we isolate the logical fact used in [Theorem 8.4](#).

Lemma A.1. *In ZF, let V be a well-orderable set of propositional variables and let T be a set of formulas over V . If every finite subset of T is satisfiable, then T is satisfiable.*

Proof. The set of finite formulas over a well-orderable alphabet is well-orderable. Enumerate it as $\langle \varphi_\alpha : \alpha < \theta \rangle$. Starting with $T_0 = T$, recursively let

$$T_{\alpha+1} = \begin{cases} T_\alpha \cup \{\varphi_\alpha\}, & \text{if this is finitely satisfiable,} \\ T_\alpha \cup \{\neg\varphi_\alpha\}, & \text{otherwise,} \end{cases}$$

and take unions at limit stages. At least one successor choice preserves finite satisfiability: if both failed, finite witnesses from the two failures would combine into a finite unsatisfiable subset of T_α . The union T^* is complete and finitely satisfiable. Set a variable p to true exactly when $p \in T^*$. Induction on formula complexity shows that every member of T^* is true under this valuation, and hence every member of T is true. $\square$

B Reproduction notes

The archive contains:

- `article.tex`: the complete source of this report;
- `article.pdf`: the compiled report;
- `verification.py`: the finite exhaustive checks described in [Section 12](#) and the computational section;
- `README.txt`: a short build and verification guide.

To run the finite checks:

```
python3 verification.py
```

The script uses only the Python standard library. The $n = 7$ exhaustive check is the largest one and may take several seconds to a few minutes, depending on hardware.

[write] **Names in this appendix, 5 October 2026.** The list above describes the delivered archive. There the source and the PDF were named `From_Finite_Blacklists_to_Freiling_Symmetry.tex` and `.pdf`, not `article.tex` and `article.pdf`; the archive also held `verification_output.txt` and `SHA256SUMS.txt`. In this report the source is `article.tex`, the program is `code/verification.py`, its recorded output is `data/verification_output.txt`, and the README replaces `README.txt`. The checks are described in [Section 11](#), not in [Section 12](#). The program writes no file and can be run from any directory; on Windows use `py` for `python3`. The whole run took about 6 seconds at the write.

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